

Econ 30065: Microeconomics

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Oligopoly (I)

Oligopoly and Game Theory

- Oligopoly: situation with few sellers and many buyers
- In an oligopoly, each firm's profit depends on the choices (P or Q) of the other firms: *strategic interaction*
- Economists determine the outcome of oligopolistic competition by applying game theory
- As we saw, in game theory a firm's most profitable choice given the actions of its rivals is called its **best response**
- In a Nash equilibrium of an oligopoly, each firm is making a profit-maximizing choice given the choices of its rivals → NE in oligopoly is a pair of best responses to each other's strategy!

Oligopoly Pricing Game with discrete set of actions

- Suppose Pepsi/Coke can each choose to sell their product at high or low price
- If one charges the high price but the other charges the low → most customers will go to the company that sells is at low price
- Their joint profit would be highest if they could agree to both charge the high price (yellow area)-Pareto efficiency
- What's the NE of this game?

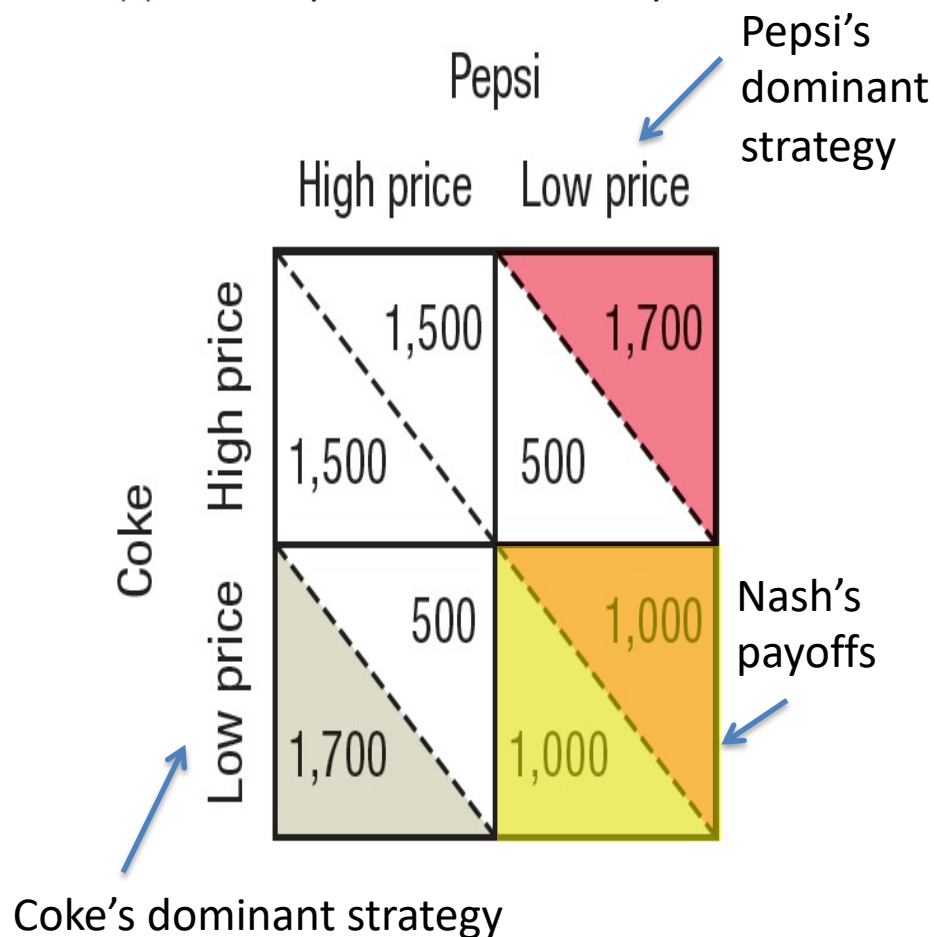
(a) Profits

		Pepsi	
		High price	Low price
Coke	High price	1,500 1,500	500 1,700
	Low price	1,700 500	1,000 1,000

Oligopoly Pricing Game

- Each firm has an incentive to undercut the price of the other one
- But if both undercut their price, both will have lower profits.
- In the end, NE will be (Pepsi's price low; Coke's price low)
- This is a general problem facing every oligopolist firm.
- This is an example of *Prisoner's Dilemma*.
- Charging the low price is a *dominant strategy* for both

(b) Best responses and Nash equilibrium



Bertrand Model

Bertrand model of oligopoly: *few* firms produce *homogeneous* (i.e identical) or *differentiated* products and *set their prices simultaneously*

Assumption: constant marginal cost

Bertrand Model with homogenous goods

- To keep things simple we study a Bertrand duopoly (Bertrand model with only two firms)
- Assumption: firms have constant marginal cost and no fixed costs
- Once firms have set their prices simultaneously, buyers observe prices and decide how much to purchase from each firm
- They purchase from firm with *lower price*
 - Each firm's most profitable choice depends on what the other does:
 - If the other firm has set a lower price, no one will buy from the first
 - If they pick the same price, demand is split in two
 - If the other firm has set a higher price, everyone will buy from the first

How to solve the game?

- One-stage game
- Infinite set of actions
- ➔ Apply procedure for *one-stage games with finely divisible choices*:
 - 1- find best response functions
 - 2- see where they cross each other: that is NE

Firms' profits

Firm 1:

$$P_1 \times Q - TC_1 \quad \text{if } P_1 < P_2$$

$$P_1 \times 0.5Q - TC_1 \quad \text{if } P_1 = P_2$$

$$0 \quad \text{if } P_1 > P_2$$

Firm 2:

$$P_2 \times Q - TC_2 \quad \text{if } P_2 < P_1$$

$$P_2 \times 0.5Q - TC_2 \quad \text{if } P_2 = P_1$$

$$0 \quad \text{if } P_2 > P_1$$

Where Q is the total market demand

→ each firm wants to set the lowest price to maximize profits!

Firms' best response function

What is the best response function of each firm?

$$P_1 = \begin{cases} MC_1 & \text{if } P_2 \leq MC_1 \\ P_2 - 0.01\$ & \text{if } MC_1 < P_2 \leq P_1^M \\ P_1^M & \text{if } P_2 > P_1^M \end{cases}$$

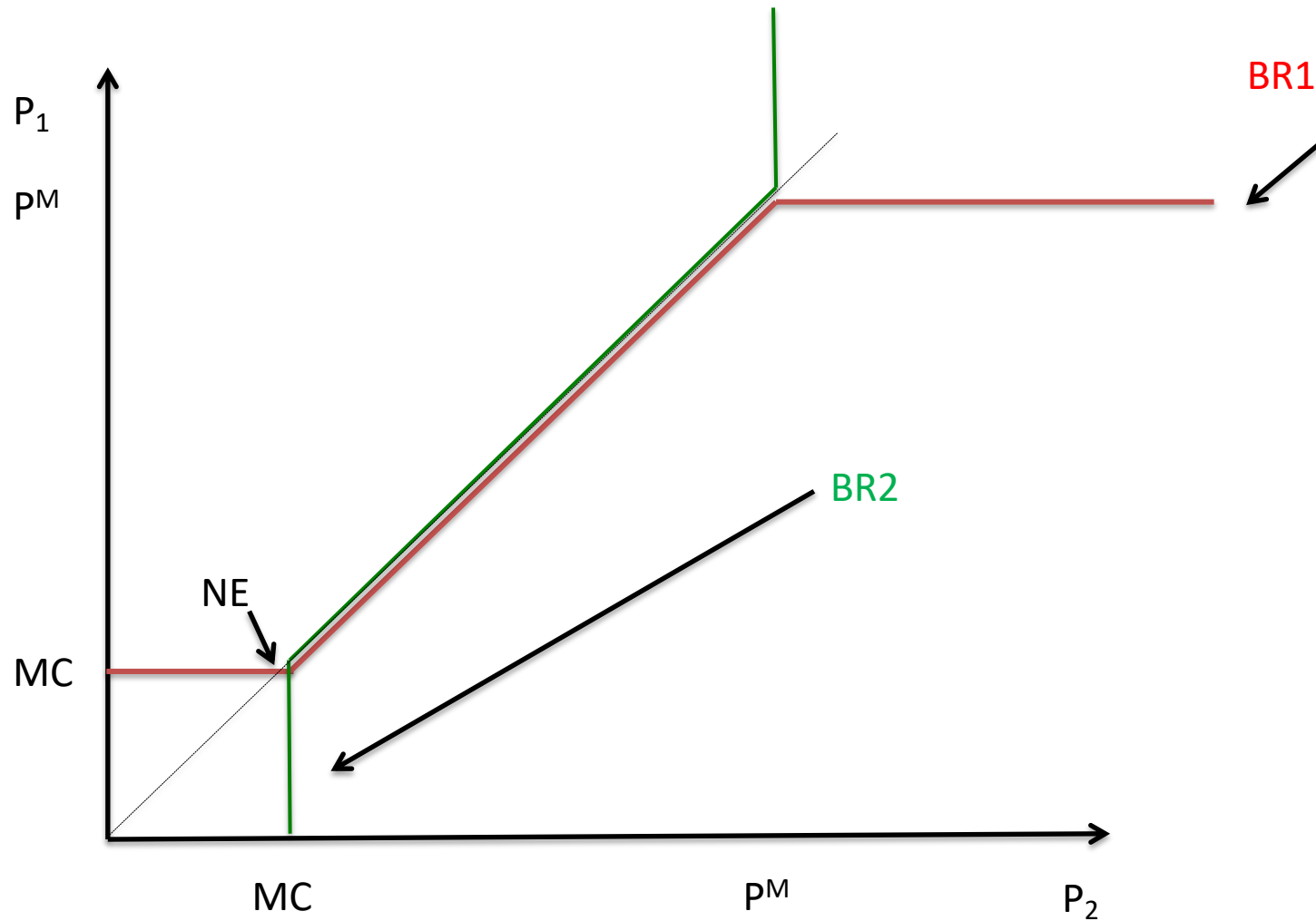
$$P_2 = \begin{cases} MC_2 & \text{if } P_1 \leq MC_2 \\ P_1 - 0.01\$ & \text{if } MC_2 < P_1 \leq P_2^M \\ P_2^M & \text{if } P_1 > P_2^M \end{cases}$$

Nash Equilibrium

- Look at firms' best responses:
- If firms have same MC:
 - $P_1 = P_2 = MC$ is the only stable outcome!
 - Graphically $P_1 = P_2 = MC$ is the only point where the best responses cross!
 - They split the demand in 2
- ➔ NE: (MC, MC)

- If $MC_1 > MC_2$:
- $P_1 = MC_1$ and $P_2 = MC_1 - 0.01\$$ is the only stable outcome
- all the demand would go to firm 2!
- ➔ NE: $(MC_1, MC_1 - 0.01\$)$

Best response functions



Example

Assume two firms have same $MC=40\$$ and they are a duopoly on the market. Assume that inverse market demand is given by $P=100-2Q$. If the 2 firms set their price simultaneously and independently, what is the NE of the game?

Example

1) Describe best responses:

$$P_1 = \begin{cases} 40 & \text{if } P_2 \leq 40 \\ P_2 - 0.01\$ & \text{if } 40 < P_2 \leq 70 \\ 70 & \text{if } P_2 > 70 \end{cases}$$

$$P_2 = \begin{cases} 40 & \text{if } P_1 \leq 40 \\ P_1 - 0.01\$ & \text{if } 40 < P_1 \leq 70 \\ 70 & \text{if } P_1 > 70 \end{cases}$$

2) Find NE: $P_1 = P_2 = MC = 40$

3) They split the demand $30/2 = 15$ each!

Nash Equilibrium and Bertrand Duopoly

- Bertrand outcome is the same as the perfectly competitive outcome ($P=MC$)
- Implies that welfare losses due to market power are limited to monopoly markets
 - Overly optimistic
 - Some of the model's assumptions may be at odds with reality:
 - Capacity limits! ➔ look at quantity decisions: Cournot Model